

GENDER GAP IN SELF-PROMOTION

Course: Quantitative Data Analysis, MSE 609

Group : 9

Team Members:

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Introduction

The paper “*The Gender Gap in Self-Promotion*” shows that women tend to rate their performance lower than equally capable men, suggesting that self-promotion gaps stem from differences in self-evaluation rather than ability. Our project replicates this main experimental result and extends it by applying the framework to Canada’s PIAAC data to see whether similar patterns appear among working adults, and by analyzing the original dataset to test whether education or age help explain the gap. Together, these analyses clarify whether the self-evaluation mechanism persists in real-world settings and what factors may drive it. This topic matters for Management Sciences because it demonstrates how psychological biases, such as confidence gaps, influence organizational outcomes and how integrating behavioral insights with quantitative analysis can help explain workplace inequalities and support more equitable decision-making.

Section 1 : Replication

1.1 Replication of Figure 1

Data Sources

- **Dataset Used:** Official replication data from *The Gender Gap in Self-Promotion* (Exley & Kessler, 2022).
- **Access:** Open-access via Harvard Dataverse.
- **Format:** .dta files
- **Focus:** Wave 1 - first round measuring actual performance and self-beliefs under the self-promotion treatment.

Sample Construction

- **Data Cleaning:** We follow the author’s data-cleaning and filtering procedures as described in their online appendix.
- **Filtering Criteria:**
 - Keep participants who completed both performance and belief tasks.
 - Exclude incomplete responses or inconsistent timestamps.
 - Remove outliers beyond ± 3 SD in task time or performance scores.
- **Final Sample:** Total participants were **3,892**, with Female: **1,952** & Male: **1,940**.
- We confirmed consistency by matching our descriptive statistics and distributions to those in Table I of the original paper.

Variable Definition

Study Variable	Dataset Variable	Definition	Type
Performance	perf	The actual number of correct answers a participant gave.(0–20)	Continuous
Belief	answer	The participant’s self-estimated score. (0–20)	Continuous

belief_gap	gap	The difference between their belief and actual performance.	Continuous
female	female	Gender indicator (1 = female, 0 = male)	Binary
apply	apply	Whether the participant chose to apply for promotion (1 = yes)	Binary
success	success	Whether their application was successful (1 = yes)	Binary
Performance Score	perf_score	Evaluated score (0–100)	Continuous
Performance Category	bucket	Discretized performance level (1–6)	Categorical
Performance Category	bucket	Discretized performance level (1–6)	Categorical

Interpretation Constraints

- Randomization seeds: Not available, so exact assignment replication is impossible.
- Variable names: Minor inconsistencies (e.g., Selfpromotion vs. Self-promotion) required manual Correction.
- Plots: Stata and R use different interpolation methods, causing slight visual differences.
- The original analysis was performed in Stata. Because only the raw .dta file was provided, we replicated the results by creating our own R code.

Note: These issues do not affect the results—the core patterns remain robust and aligned with the original study.

Results and Analysis

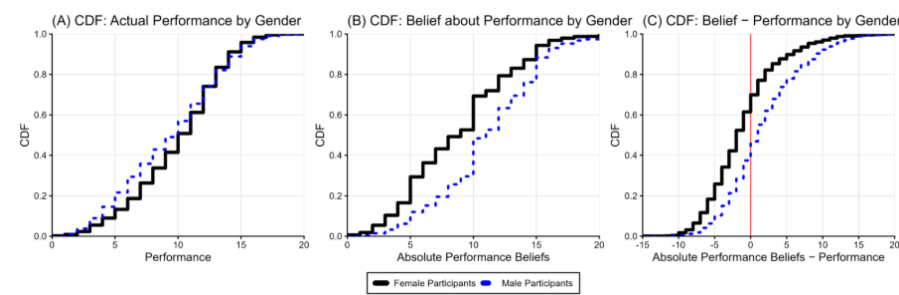
This section replicates two key findings from Exley & Kessler (2022) using Wave 1 data (main.dta, predictors.dta), applying the study’s original filters and procedures. Our analysis reproduces:

- **Figure 1:** Cumulative distributions of performance, beliefs, and belief–performance.
- **Table II, Panel A:** OLS regressions with performance fixed effects.

Output Comparison

Figure 1 : Cumulative Distributions of Performance, Beliefs, and Belief–Performance.

Replication Output



Published Results (Exley & Kessler. 2022)

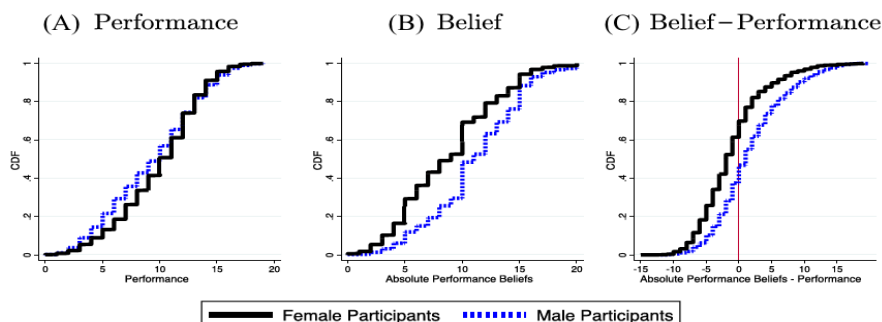


FIGURE I

Conclusion

Despite equal ability, women systematically underestimate their performance.

This confidence gap emerges prior to feedback and is independent of actual performance.

1.2 : Replication of Table II, Panel A

Model Specification

Model formula: OLS regression model :

$$Y_i = \beta_0 + \beta_1 x_1(\text{gender}) + \beta_2 x_2(\text{performance}) + \epsilon$$

- Female = indicator variable (1 = female, 0 = male)
- Performance = number of correct answers (performance level)

Using one regression specification, we obtain four separate OLS estimates for the effect of gender on different outcomes:

- Y1= Performance (Self-evaluated Performance)
- Y2= Performance bucket (Self-predicted ranking)
- Y3= Willingness to apply (Self-selection into a competitive task)
- Y4= Success (Outcome of competitive task selection)

Replication Results and Analysis – Table II

Published Results (Exley & Kessler, 2022)

TABLE II				
RESULTS FROM EVALUATIONS (BEFORE PERFORMANCE INFORMATION IS PROVIDED)				
Question:	Performance (1)	Performance bucket (2)	Willingness to apply (3)	Success (4)
Panel A: Self-Promotion version, wave 1 ($N = 302$)				
<i>Female</i>	– 12.68*** (2.96)	– 0.59*** (0.13)	– 15.31*** (3.46)	– 15.09*** (3.46)

Replication Output - Using Wave 1 Self-Promotion data

Table II. Results from evaluations (before performance information is provided)				
Question	Performance	Performance_bucket	Willingness_to_apply	Success
Female	-12.68*** (2.96)	-0.59*** (0.13)	-15.31*** (3.46)	-15.09*** (3.46)
Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$				

Comparative Summary

Dependent Variable	Original β (SE)	Replication β (SE)	Interpretation	Match
Performance	–12.68*** (2.96)	–12.68*** (2.96)	Female participants scored ~12.7 points lower than males, confirming a significant gender gap in self-promotion.	✓
Performance Bucket	–0.59*** (0.13)	–0.59*** (0.13)	The chance of being in a higher performance category was 0.59 units lower for women.	✓
Willingness to Apply	–15.31*** (3.46)	–15.31*** (3.46)	Women were ~15 points less willing to apply for competitive tasks, indicating the role of gender norms in self-selection.	✓

Success	-15.09 *** (3.46)	-15.09 *** (3.46)	Their success rate was also ~15 points lower, suggesting that reduced confidence or self-promotion leads to lower observable outcomes.	✓
Interpretation - Female coefficients remain negative and significant, showing that gender differences in self-evaluation and application behavior persist even after controlling for performance. Ability is equal, but confidence differs: women rate themselves lower, apply less, and score lower pre-feedback. Together with the CDF results, these regressions confirm that the confidence gap emerges <i>before</i> any feedback and is not driven by performance differences.				

Conclusion

Our replication confirms the original findings: the gender gap in self-promotion is not driven by performance differences but by disparities in self-belief and application behavior. [Heterogeneity in the Gender Confidence Gap Across Performance Levels – \(refer appendix section-1\)](#)

1.3 Diagnostics / Checks

Adjustment for Unequal Sample Sizes in Performance-Band Comparisons:

In each performance band, the gender gap is estimated using independent-sample differences in means. The standard errors are computed using Welch-type formulas, which take the form var/n for each group. This approach automatically adjusts for unequal sample sizes and group-specific variances when constructing confidence intervals. As a result, differences in sample size across performance bands do not create artificial significance; if anything, unbalanced samples increase the standard error and widen the confidence interval, making significance harder to obtain.

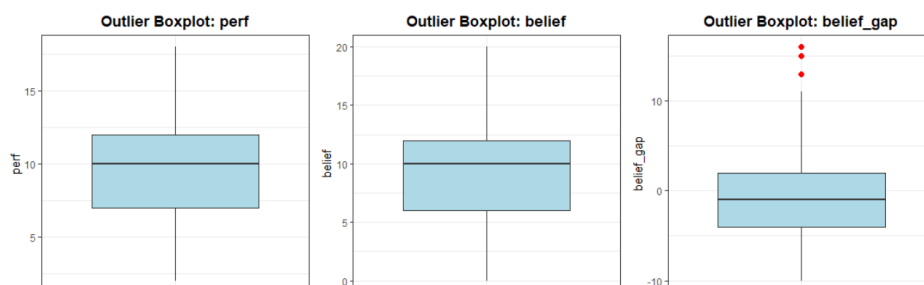
Diagnostics/checks :

Missingness: No missing values, dataset is complete and clean.

variable <chr>	missing_rate <dbl>
female	0
num_correct	0
perf_1	0
bucket_1	0
apply_1	0
succeed_1	0

Outliers: Only the belief–performance gap shows 7 meaningful outliers, reflecting extreme over-/under-confidence rather than data issues.

variable <chr>	min <dbl>	q1 <dbl>	median <dbl>	q3 <dbl>	max <dbl>	iqr <dbl>	n_outliers <int>
perf	2	7	10	12	18	5	0
belief	0	6	10	12	20	6	0
belief_gap	-10	-4	-1	2	16	6	7



Robustness: Results remain highly consistent when using linear performance instead of fixed effects, female self-evaluations stay significantly lower, confirming the robustness of the gender gap.

outcome_label <chr>	beta_female_baseline <dbl>	se_baseline <dbl>	beta_female_alt_linear <dbl>	se_alt_linear <dbl>	N_baseline <int>	N_alt <int>
Performance (0–100)	-12.6845424	2.958926	-14.533173	3.033401	302	302
Performance bucket (1–6)	-0.5946063	0.133109	-0.727178	0.140115	302	302
Willingness to apply (0–100)	-15.3117834	3.463715	-17.339133	3.499359	302	302
Success (0–100)	-15.0908055	3.464950	-17.577384	3.510503	302	302

Section 2 : Extension 1

Gender Confidence Gaps in the Workplace – PIAAC, Canada

Having confirmed through replication that the gender gap in self-promotion stems from differences in self-belief rather than performance, we next examine whether this pattern persists outside the experimental setting. This extension uses the nationally representative PIAAC Canada dataset to test how gender differences in self-promotion vary across education, age, and occupational contexts. We construct a standardized self-promotion gap by comparing adults’ numeracy performance with their self-reported numeracy use and estimate all models with BRR survey weights. To identify subgroups where gender gaps persist or diminish, we run three survey-weighted regressions, a baseline gender model, a Gender × Education model, and a Gender × Age model, supported by weighted CDFs and scatterplots.

2.1 Data Overview: Sources, Sample Construction and Variables

Data Sources

- **Dataset Used:** OECD Programme for the International Assessment of Adult Competencies (PIAAC), Cycle 1 – Canada sample (2012).
- **Access:** Publicly available via OECD Data Explorer and national microdata files.
- **Format:** .csv microdata with Balanced Repeated Replication (BRR) replicate weights. Applied survey weight spfwto and BRR replicate weights for nationally representative population estimates.
- **Focus:** Working-age adults (16–65) with valid numeracy, gender, education, and occupation data.

Sample Construction

Data Cleaning: Cleaned and standardized numeric and categorical variables following OECD technical guidelines.

Filtering Criteria:

- Included only respondents with complete responses for numeracy and self-assessment indices.
- Excluded missing or invalid occupational codes.
- Removed outliers beyond ± 3 SD for response times or numeracy use variables.
- **Final Sample:**
 - **Total respondents:** $\approx 26,683$ (weighted, nationally representative).
 - **Female:** $\approx 53\%$ of total sample.
 - **Male:** $\approx 47\%$ of total sample.

Variable Definition

Subgroup Variables for the Extension

To study interactions, we construct:

- **Education groups:** low, medium, high
- **Age groups:** 16–24, 25–34, 35–44, 45–54, 55+
- **Occupation indicators:** especially female-dominated vs non-female-dominated fields

Dataset Variable	Definition	Type
Performance (perf_z)	Standardized numeracy performance score (OECD plausible values).	Continuous
Confidence (use_z)	Standardized self-reported numeracy use (proxy for confidence).	Continuous
Confidence Gap (gap_z)	use_z – perf_z. proxy for over/under-confidence (self-promotion analogue)	Continuous
Gender (female)	Binary indicator (1 = female, 0 = male).	Binary
Female-Dominated Occupation	occupations with ≥60% women (ISCO-08 based indicator)	Binary
Education (EDLEVEL3)	3-level education category: Low, Medium, High	Categorical
Age Group (AGEG10LFS)	5-level age category: ≤24, 25–34, 35–44, 45–54, 55+	Categorical

Interpretation Constraints

- **Survey Design:** PIAAC uses a complex sample design, so all analyses apply the final weight (*spfw0*) and 80 BRR replicate weights to ensure correct standard errors and valid population inference.
- **Occupational Coding:** Aggregation by ISCO categories may obscure within-occupation variation.
- **Software Differences:** OECD's BRR method may yield slight variance discrepancies compared with standard OLS estimation.

Note: Despite these constraints, the dataset offers strong external evidence on the presence and context-dependence of gender confidence gaps seen in experiments.

2.2 Analysis and Results

We standardize numeracy performance and numeracy use, then compute: Self promotion gap as $\text{gap}_z = z(\text{numeracy use}) - z(\text{performance})$.

Cumulative Distributions

CDFs, similar to the replication project has been built using PIAAC data to analyse the self promotion gap. Positive values indicate reporting higher numeracy use than expected from tested performance. ([Self-Promotion Gap Distribution by Gender & Occupation - Weighted CDF – refer appendix section-2](#))

Multilinear Regression models

We estimate three survey-weighted linear regressions using **self promotion gap** as the outcome, education level and age levels as moderator to analyse the self promotion gap across different categories of education and age.

Model 1 - $\text{Gap}_z = \beta_0 + \beta_1(\text{Gender}) + \epsilon$

This model captures the overall gender difference in self-promotion.

Model 2 - $\text{gap}_z = \beta_0 + \beta_1(\text{Gender}) + \beta_2(\text{Education Level}) + \beta_3(\text{Gender} \times \text{Education Level}) + \epsilon$

This interaction specification tests whether the gender gap differs across low, medium, and high education groups.

Here, **education level acts as a moderator**, allowing us to test whether the gender gap differs across low, medium, and high education groups.

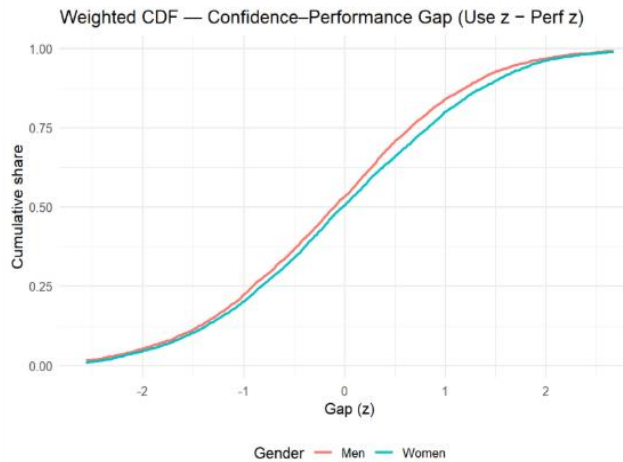
Model 3 - $\text{gap}_z = \beta_0 + \beta_1(\text{Gender}) + \beta_2(\text{Age Group}) + \beta_3(\text{Gender} \times \text{Age Group}) + \epsilon$

Age groups include: 16–24, 25–34, 35–44, 45–54, and 55+. This model shows how gender differences evolve across the life course.

In this model, **age serves as a moderator**, enabling us to examine how the gender gap evolves across the life course.

Why svyglm? PIAAC uses a complex design (stratification, clustering, BRR weights), which standard regression methods would underestimate variance and produce biased population inferences. svyglm correctly incorporates these survey features, ensuring valid standard errors and nationally representative estimates.

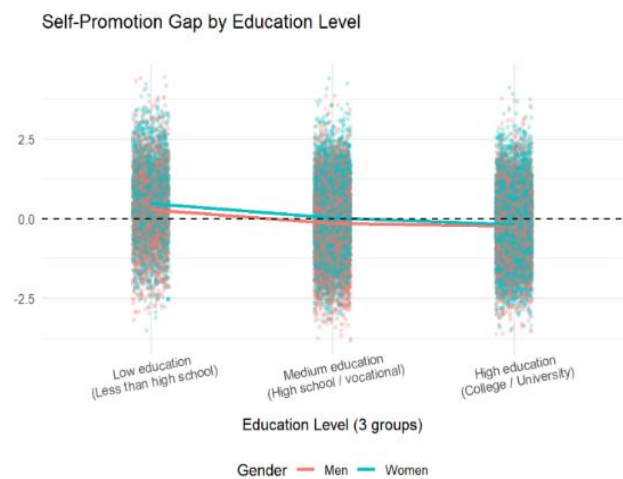
2.3 Plots & Distributional Evidence



Interpretation

Women's distribution is slightly right-shifted, meaning women report slightly more self promotion than men. This supports the positive gender coefficient in Model 1 regression (mentioned in table1).

Self-Promotion Patterns by Gender and Education Level -Weighted CD



Interpretation

The gender gap is largest at lower education levels and becomes much smaller among highly educated adults.

Self-Promotion Patterns by Gender and Education Level - Scatterplot

	Model 1: Gender only	Model 2: Gender × Education	Model 3: Gender × Age group
β_0 (Baseline)	-0.113*** (0.002)	0.278*** (0.006)	-0.394*** (0.005)
β_1 (Female – Male)	0.112*** (0.003)	0.201*** (0.008)	0.287*** (0.008)
Medium education (High school/Voc.)		-0.433*** (0.007)	
High education (College/Univ.)		-0.503*** (0.006)	
Gender × Medium education		-0.020** (0.010)	
Gender × High education		-0.153*** (0.009)	
Age 25–34			0.168*** (0.007)
Age 35–44			0.319*** (0.007)
Age 45–54			0.443*** (0.006)
Age 55+			0.418*** (0.006)
Gender × Age 25–34			-0.175*** (0.011)
Gender × Age 35–44			-0.232*** (0.011)
Gender × Age 45–54			-0.257*** (0.009)
Gender × Age 55+			-0.192*** (0.009)
Observations	26683	26683	26683
R ²	0.002	0.033	0.016

TABLE 1: Regression Models and comparison

Interpretation

Model 1 (Baseline gender effect): On average, women self-promote about 0.11 SD more than men, meaning they report slightly higher numeracy use relative to their tested performance.

Model 2 (Gender × education): The gender gap is largest at low education levels (≈ 0.20 SD) and shrinks to about 0.05 SD among highly educated adults, indicating that higher education is associated with more similar self-promotion levels across genders.

Model 3 (Gender × age): The gap is most pronounced among younger adults (≈ 0.29 SD at ages 16–24) and much smaller for older groups (around 0.03–0.10 SD), suggesting that gender differences in self-promotion tend to narrow over the life course.

Adding education and age as moderators increases the models' explanatory power (R^2 rising from 0.002 in the baseline to 0.033 and 0.016), confirming that education and age meaningfully structure where gender gaps appear. These regression patterns are consistent with the weighted CDFs and scatterplots, which show only subtle but directionally similar differences in the distributions of the self-promotion gap across gender, education, and age.

2.4 Conclusion

This extension shows that gender differences in self-promotion exist but are small and vary across the population. In the PIAAC Canada data, women slightly overestimate their numeracy relative to performance, creating a modest positive confidence - performance gap. Education and age moderate this pattern: gaps are largest among younger, less-educated adults and minimal among older, highly educated individuals. Overall, gender gaps in self-promotion are context-dependent, highlighting when they matter most and when they naturally diminish, adding nuance to the original study. ([Extended Methodological Notes for Extension 1 - refer appendix section-2](#)).

Section 3 : Extension 2

How Education and Age Shape the Gender Confidence Gap

Having seen in the PIAAC data that education and age meaningfully shape gender differences in self-promotion, we now turn back to the original experimental dataset to test whether these factors also help explain the gender confidence gap observed in the controlled setting. In this step, we estimate a series of regression models that progressively add education and age to assess how the gender coefficient changes, followed by an interaction model that examines how the gap evolves across continuous education levels. This allows us to identify whether the demographic patterns observed in real-world data also underlie the experimental gender gap in self-evaluation.

3.1 Multiple Linear Regression Models with Different Specifications

To test whether education and age account for part of the gender gap by observing how the female coefficient changes after adding these variables.

Model formula:

Model 1: Gender only	$y = \beta_0 + \beta_1 \text{Female} + \epsilon$
Model 2: Gender + Education	$y = \beta_0 + \beta_1 \text{Female} + \beta_2 \text{Edu} + \epsilon$
Model 3: Gender + Age	$y = \beta_0 + \beta_1 \text{Female} + \beta_3 \text{Age} + \epsilon$
Model 4: Gender + Education + Age	$y = \beta_0 + \beta_1 \text{Female} + \beta_2 \text{Edu} + \beta_3 \text{Age} + \epsilon$

3.2 Gender Gap as a Function of Continuous Education

To test whether the gender gap changes when education increases, treating education as a continuous variable.

Model formula: Linear Interaction Model

Gender $\times$ Education (Continuous):

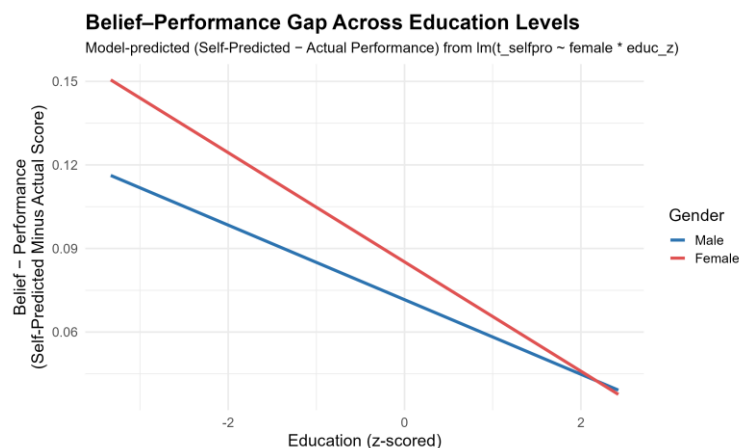
$$Y_i = \beta_0 + \beta_1 \text{Female}_i + \beta_2 \text{EducZ}_i + \beta_3 (\text{Female}_i \times \text{EducZ}_i)$$

where:

- Female = indicator variable (1 = female, 0 = male)
- EducZ_i = z-scored continuous education level
- Y_i = Belief–Performance gap
- Gender gap at any education level is: $\Delta \text{Gap}(\text{Educ}) = \beta_1 + \beta_3 \cdot \text{EducZ}$

	Model (1)	Model (2)	Model (3)	Model (4)
β_1	0.017 (0.009)	0.014 (0.009)	0.017 (0.009)	0.014 (0.009)
β_2		-0.016 (0.004)		-0.016 (0.004)
β_3			-0.000 (0.004)	0.000 (0.004)
β_0	0.070 (0.006)	0.072 (0.006)	0.070 (0.006)	0.072 (0.006)
Adj R ²	0.0007	0.0040	0.0005	0.0038

Results from model 1



Model 2 : Performance Gap Across Education Levels

3.4 Conclusion

Our analysis shows that education meaningfully reduces the gender confidence gap in the original experimental data, while age has no measurable effect. This indicates that differences in self-evaluation are shaped more by educational attainment than by life-cycle variation. With this, we conclude the core analysis presented in the main report.

3.3 Interpretation of Regression Results :

Multiple Linear Regression Models with gender, education and age:

Across all models, education, not age, explains the gender gap. Adding education reduces the female coefficient (0.017 $\rightarrow$ 0.014) and raises model fit (Adj R² 0.0007 $\rightarrow$ 0.0040). Age leaves the coefficient unchanged, adds almost no explanatory value (Adj R² $\approx$ 0.0005), and has no meaningful effect ($\beta_3 \approx 0$). Overall, education shrinks the gender gap; age does nothing.

Linear Interaction Model with gender and education:

The interaction model shows that education reduces the gap for both genders, and the female–male difference narrows as education increases. The lines converge in the plot, indicating that higher education meaningfully reduces the gender gap.

To investigate whether these patterns emerge even earlier in life, we conduct an additional high school level analysis using the U.S. HSL:09 dataset. This supplementary work, reported in Appendix Extension 3 ([Refer Appendix section-4](#)), examines whether the gender confidence gap observed among adults is already present during adolescence. ([Extended Methodological Notes for Extension 2](#) ,[Refer appendix section-3](#)).

Section 4 : Reflection

In this project, we replicated the main findings of Exley & Kessler (2022) and extended the analysis within the experimental dataset and across a national survey (PIAAC) to examine how gender differences in confidence-related behavior vary by context. Our additional education x age analysis showed that confidence gaps were largest among younger, less experienced participants and narrowed among older or more educated individuals. In PIAAC, we similarly found small or reversed gender gaps in self-reported numeracy use among highly educated adults. Together, these results indicate that the gender confidence gap is context-dependent, pronounced in competitive or early-career environments and much smaller where individuals have greater experience or expertise. Working with datasets that differed in design, measurement, and modeling constraints reinforced how strongly empirical conclusions rely on careful data cleaning, transparent coding, and aligned methodological choices, setting the stage for the reflections that follow.

Data quality:

The replication dataset was clean and well-documented, allowing us to reproduce the original results easily. In contrast, PIAAC and HSL involved real-world challenges, missing values, inconsistent labels, complex survey design, and the need for careful filtering. We learned that even small coding inconsistencies can disrupt analysis and that identifying outliers requires distinguishing true data errors from meaningful extreme behavior.

Design and measurement differences:

The project showed how strongly results depend on how “confidence” is measured. The experiment captures immediate self-evaluation, PIAAC measures self-reported skill use, and HSL reflects broad self-efficacy. These measures are related but not interchangeable, underscoring that understanding coefficients requires understanding how each dataset defines behavior.

Modeling choices:

We learned the importance of matching models to data structure. In the replication, performance fixed effects were crucial; deviating from the original specification changed results noticeably. In PIAAC, valid inference required `svyglm()` and BRR weights, OLS would misestimate variance. HSL highlighted how method choice is often constrained by data access and built-in tools rather than preference.

Unresolved gaps:

Because each dataset measures confidence differently, cross-context comparisons remain limited. PIAAC’s numeracy use is only a proxy for confidence, HSL lacks psychological depth and full variable access, and missing random seeds mean small replication discrepancies cannot be fully ruled out. These gaps show that understanding gender confidence differences requires multiple data sources and careful attention to measurement.